Bridging the Data Gap: Few-Shot Anatomical Segmentation in Abdominal MRI via the Reptile Meta-Learning Algorithm

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Abstract. Abdominal magnetic resonance imaging (MRI) segmentation is clinically important but inherently limited by the scarcity of annotated data, particularly for newly defined anatomical structures. Under such conditions, conventional supervised models often fail to generalise reliably across unseen structures. This work evaluates whether first-order meta-learning can improve adaptation in this regime. A Reptile-based framework is used to learn an initialisation for a compact U-Net that can be efficiently adapted to new anatomical segmentation tasks using only 1–5 labelled examples. The approach is assessed against supervised training under controlled few-shot conditions, including unseen tasks, varying support set sizes, and multiple random seeds to account for sampling variability. Results show that meta-learned initialisation provides a consistent advantage in the most data-scarce setting, with improved performance at 1-shot and a modest benefit at 3-shot on unseen test tasks, although this advantage is not sustained at 5-shot. The overall performance remains limited, reflecting both the intrinsic complexity of abdominal MRI segmentation and the restricted diversity of the meta-training distribution. These findings position first-order meta-learning as a principled, data-efficient alternative to conventional supervised approaches for few-shot anatomical segmentation, while highlighting the critical role of task diversity and optimisation in achieving robust performance.

Keywords: Few-shot learning, reptile algorithm, image segmentation, abdominal MRI.

1 Introduction

Magnetic resonance imaging (MRI) has revolutionised clinical diagnostics over the last three decades, offering superior soft-tissue contrast, absence of ionising radiation and multiplanar capability [1] [2]. Anatomical segmentation is a critical prerequisite for quantitative MR image analysis, enabling accurate diagnosis and therapy planning. However, abdominal MR segmentation remains challenging due to anatomical complexity and multiple organs-at-risk. It still relies heavily on manual segmentation, which is labour-intensive and prone to inter-observer variability [3]. These limitations have driven interest in automated approaches, particularly deep learning methods.

Deep convolutional neural networks, notably encoder-decoder architectures such as U-Net [4] and its variants, including Attention U-Net [5] and TransUNet [6], have achieved strong performance in fully supervised settings. However, their application to abdominal MRI, especially for intricate structures such as the stomach and duodenum [7], remains limited by the need for vast repositories of densely labelled data and poor generalisation to unseen anatomies [3]. Under data-scarce conditions, such models are susceptible to overfitting due to insufficient training constraints [8].

Few-shot learning (FSL) addresses this limitation by enabling rapid generalisation to new tasks from a small support set of k labelled examples. Within FSL, two principal paradigms have emerged: metric-based approaches, such as Prototypical Networks [9], which perform classification by computing distances to class prototypes in a learned embedding space; and optimisation-based meta-learning methods, which learn parameter initialisation that enables rapid adaptation via a few gradient descent steps [10]. Model-Agnostic Meta-Learning (MAML) [10] formalises this objective through a bi-level optimisation, but its requirement for second-order derivatives introduces substantial memory overhead; on a 12 GB GPU, memory consumption scales linearly with the number of inner-loop steps, becoming prohibitive at approximately 16 steps for a four-layer convolutional network [11].

Reptile [12] offers a computationally efficient first-order alternative: it interpolates the meta-parameters towards the task-adapted parameters, avoiding the need to differentiate through the optimisation trajectory. Taylor expansion analysis reveals that the Reptile update implicitly maximises the inner product between gradients of distinct mini-batches drawn from the same task, a quantity directly correlated with generalisation [12]. Empirically, Reptile achieves performance competitive with both MAML and its first-order variant (FO-MAML) on standard few-shot classification benchmarks, whilst requiring considerably less computation. This property makes Reptile particularly appealing to the resource-constrained medical imaging community, given the high inter-patient anatomical variability in abdominal MRI.

Recent studies validate gradient-based meta-learning for medical segmentation. Oliveira et al. [13] demonstrate that such methods generally outperform prototypical approaches when label distributions shift significantly between training and test tasks. Khadka et al. [14] improve MAML stability via implicit gradients, Feng et al. [15] incorporate spatial priors and prototype similarity for enhanced few-shot segmentation on abdominal MRI datasets and Farshad et al. [16] utilise cross-task meta-training to improve adaptation to novel organs. Collectively, these findings motivate the use of meta-learning for anatomical segmentation in abdominal MRI, where organs exhibit anatomical and contrast variability across patients and acquisition protocols.

This study evaluates a Reptile-based meta-learning framework for few-shot anatomical segmentation in abdominal MRI under extreme data scarcity within a controlled environment. It compares meta-learned initialisation to a supervised baseline across varying support sizes (k = 1, 3, 5). Performance is measured by Dice Similarity Coefficient (DSC) and Intersection over Union (IoU), assessing model stability and adaptation to unseen structures. The work investigates whether first-order meta-learning provides an efficient alternative to conventional approaches when expert annotations are critically limited.

2 Methods

2.1 Data Preparation and Model Architecture

The study utilises 2D slices from an adapted version of the Cross-institution Male Pelvic Structures Dataset [17], partitioned into task-specific dictionaries to prevent data leakage. To standardise the input, all MR slices and masks are resized to 128x128 pixels. A custom PyTorch dataset handles grayscale conversion and mask binarisation to [0, 1]. The base model is a compact U-Net [4] with an encoder-decoder architecture and skip connections. This version uses 16 initial feature channels, batch normalisation, and a dropout rate of 0.1 to prevent overfitting during the k-shot adaptation phase.

2.2 Reptile Meta-Learning Framework

To identify an optimal weight initialisation θ, a batched variant of the Reptile algorithm is implemented, as proposed by Nichol et al. (2018) [12]. At each meta-training iteration, a batch of $M = 4$ tasks are sampled. For each task, the model parameters are independently adapted using $k = 5$ gradient descent steps on the support set to find task-specific weights $\tilde{\theta}$. The global meta-parameters are then updated towards the average of these adapted parameters:

$$\theta \leftarrow \theta + \epsilon\left(\frac{1}{M}\sum_{i=1}^{M}\tilde{\theta}_i - \theta\right) \tag{1}$$

where $\epsilon = 0.1$ is the meta-learning rate. This batching strategy is chosen to improve optimisation stability across diverse anatomical structures.

2.3 Experimental Design and Evaluation

Experiments were conducted in a Google Colab environment (NVIDIA T4 GPU). To ensure statistical robustness, the model was evaluated across three random seeds (42, 123, and 999), and the mean performance was reported. Due to computational constraints, k-shot adaptation was evaluated for k = 1, 3, and 5, where the model was fine-tuned on support sets and tested on separate query sets. Results were compared to a supervised baseline trained from scratch on the same support data using 40 fine-tuning steps, whereas Reptile used 5 inner-loop adaptation steps. This establishes a conservative comparison; the baseline is given more adaptation budget.

Performance was quantified using Dice Similarity Coefficient (DSC) and Intersection over Union (IoU):

$$DSC = \frac{2|X \cap Y|}{|X| + |Y| + \delta} \quad , \quad IoU = \frac{|X \cap Y| + \delta}{|X \cup Y| + \delta} \tag{2,3}$$

where X is the prediction, Y is the ground truth, and $\delta = 10^{-6}$. We additionally report DSC (non-empty), which calculates the DSC exclusively on slices where the target structure is present ($|Y| > 0$) to avoid metric inflation from empty background slices.

3 Results

3.1 Quantitative Comparison

Evaluation of mean DSC (non-empty) and IoU across k-shot = 1, 3, 5 for the validation set (Tasks 4 and 6) and the test set (Tasks 1 and 8) reveals distinct performance trends between the Reptile-initialised framework and the supervised baseline (Figure 1).

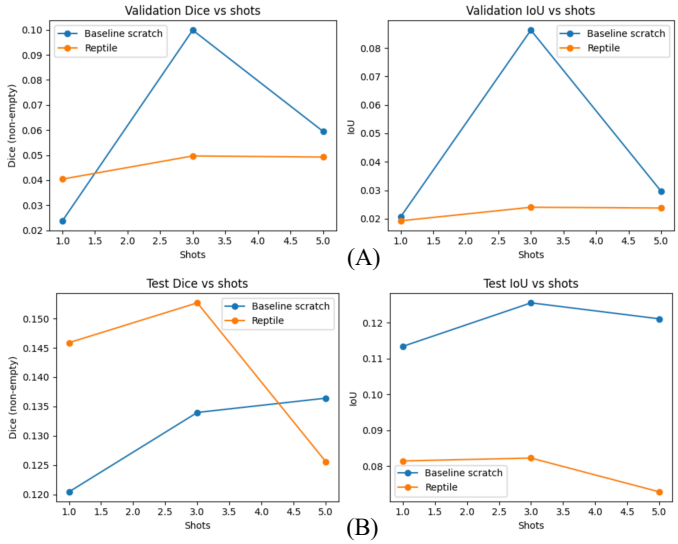


Figure 1. Mean segmentation performance vs k-shot (k = 1, 3, 5). Left: DSC (non-empty). Right: IoU. Results compare supervised baseline and Reptile across validation sets (A) and test sets (B).

Dice Similarity Coefficient (non-empty). On the validation set, Reptile outperforms baseline at 1-shot (0.0404 vs. 0.0238), consistent with the expected advantage of a task-informed initialisation under the extreme data scarcity. However, at 3-shot, the baseline achieves higher DSC (0.0999 vs 0.0497), suggesting that the 200-iteration meta-training regime may lack the stability to outperform standard fine-tuning as support data increases. The baseline's decline from 3-shot to 5-shot (0.0999 to 0.0595) reflects instability typical to training-from-scratch under limited data, whereas Reptile remains comparatively stable (0.0497 to 0.0492). In contrast, test performance favours Reptile under few-shot conditions: 0.1459 vs 0.1204 at 1-shot and 0.1527 vs 0.1340 at 3-shot. This indicates improved generalisation to unseen tasks when very limited labelled data is available. This advantage is not sustained at 5-shot, where the baseline marginally outperforms Reptile (0.1364 vs 0.1256).

Intersection over Union. On the validation set, the baseline demonstrates a strong dependence on the number of shots, increasing from 0.0206 at 1-shot to a peak of 0.0863 at 3-shot, before declining to 0.0296 at 5-shot. Reptile exhibits consistently low and relatively flat performance across all shot settings (0.0193, 0.0240, and 0.0238 for 1-, 3-, and 5-shot, respectively), suggesting limited sensitivity to additional support examples. On the test set, the baseline consistently outperforms Reptile across all shot

configurations, achieving IoU values 0.1134, 0.1255 and 0.1211 for 1-, 3- and 5-shot, respectively, compared to 0.0814, 0.0822 and 0.0728 for Reptile. These results indicate that, while Reptile may offer benefits in DSC under extremely low-data conditions, it does not translate into improved overlap quality as measured by IoU, particularly when evaluated on unseen tasks. The DSC-IoU divergence indicates that Reptile improves localisation but lacks boundary precision, producing spatially aligned predictions with diffuse or noisy edges.

Stability Analysis. Table 1 reports the mean and standard deviation of DSC (non-empty) across seeds on test tasks. The standard deviations are large relative to the means for both methods, confirming that performance is highly sensitive to the particular support examples selected, a well-documented characteristic of few-shot segmentation at very low shot counts [18].

Table 1. Mean ± standard deviation of DSC (non-empty) across seeds on test tasks

k-Shot	Baseline DSC (non-empty)	Reptile DSC (non-empty)
1	0.1204 ± 0.1200	0.1459 ± 0.1390
3	0.1340 ± 0.1485	0.1527 ± 0.1319
5	0.1364 ± 0.1355	0.1256 ± 0.1537

3.2 Qualitative Results

Figure 2 qualitatively compares task_1 across four MR slices. The baseline predicts near-complete foreground, indicating segmentation collapse from overfitting and explaining inflated DSC values in Figure 1. Reptile (1-shot) produces spatially compact, anatomically localised predictions in approximately the correct region, demonstrating that the meta-trained initialisation encodes a meaningful structural prior from a single labelled example. At 3-shot, predictions expand to better approximate the crescent morphology of the ground truth, though background noise activations persist. The 5-shot predictions show marginal boundary refinement. Notably, qualitative performance improves in lower rows where the target structure is larger and morphologically more distinctive, suggesting that structure size is an important factor in adaptation success.

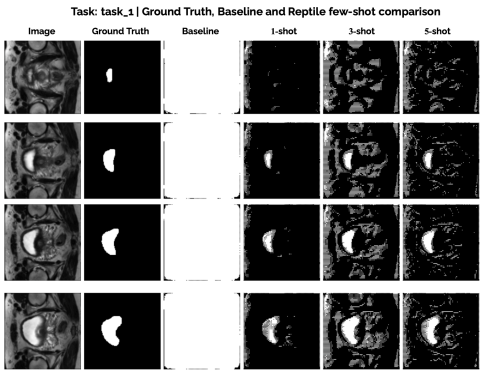


Figure 2. Qualitative comparison of baseline and Reptile predictions on task_1 across four representative MR slices under 1-, 3-, and 5-shot adaptation.

4 Discussion and Conclusion

The results demonstrate a nuanced comparison between Reptile meta-learning and supervised fine-tuning. Reptile's superior DSC (non-empty) at 1-shot and 3-shot on test tasks aligns with its objective of learning initialisations for rapid adaptation. This suggests that the meta-learner successfully captures shared anatomical features, such as landmark configurations and tissue contrast patterns, which are critical in abdominal MRI [12].

The anomalous 3-shot validation result, where the baseline exceeds Reptile (0.0999 vs 0.0497), likely reflects limitations in both optimisation and experimental design. The meta-training regime, restricted to four training tasks and 200 iterations, may not reach a stable meta-optimum. Furthermore, a fixed meta-step size of 0.1 and limited inner-loop iterations constrain convergence. Additionally, the asymmetric adaptation budget (40 steps for the baseline vs 5 for Reptile) intentionally stress-tests the meta-initialisation; under this constraint, a well-learned prior is expected to adapt efficiently with fewer updates, consistent with Reptile's objective [12].

High performance variability across seeds aligns with FSL literature, where support set selection strongly influences outcomes [18], although Reptile shows modestly improved stability at 3-shot. Task diversity is further limited by the dataset, which contains only male pelvic structures. While this restricts meta-training breadth, it reflects realistic clinical scenarios where labelled data are scarce across all categories. Lower IoU relative to the supervised baseline suggests that, despite improved localisation (DSC), boundary precision remains limited, likely due to conservative segmentation under sparse supervision [13]. These findings align with recent surveys identifying limited labeled data and high intra-class variation as significant barriers in medical FSL. Furthermore, the complexity of abdominal MRI segmentation is exacerbated by single-channel grayscale limitations and inter-slice correlation overlooks, necessitating robust meta-learning frameworks to achieve clinical-grade accuracy [19] [20].

To address these limitations, three enhancements are proposed: (a) architectural refinement, replacing skip connections with attention gates [5] to suppress irrelevant background activations common in abdominal MRI; (b) loss optimisation, transitioning to Tversky loss [21] to prioritise recall, mitigating the conservative under-segmentation bias inherent in few-shot settings; (c) transductive inference, implementing test-time fine-tuning via pseudo-labeling [8] to leverage unlabeled query distributions for improved calibration without additional training data. Despite these limitations, the findings provide credible evidence of Reptile's utility in extreme low-data regimes. Quantitative DSC advantages at 1-shot and 3-shot, alongside superior qualitative coherence, validate the meta-learned initialisation.

In conclusion, Reptile meta-learning improves few-shot abdominal MRI segmentation under extreme data scarcity, particularly in 1–3 shot settings. However, limited task diversity and optimisation constrain performance. Future work should focus on richer meta-training distributions and improved architectures to enhance generalisation and overlap accuracy.

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